

Executive Summary

Simultaneous Group-Envelope Bounds for Γ -Robust Multiple-Choice Knapsack Problems

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Many resource-allocation decisions choose exactly one option per group. Protecting their shared constraint with a Bertsimas–Sim uncertainty budget produces a robust multiple-choice knapsack problem. The classical threshold reduction is polynomial, but a large set of distinct deviations can make complete relaxation certification expensive.

After dualizing the capacity constraint of a fixed-threshold MCKP, all threshold-dependent baselines cancel. Between consecutive deviations in a group, the remaining envelope is the maximum of one constant and one common-slope line. Prefix and suffix maxima with range accumulation evaluate one multiplier over all B thresholds in $O(B + K \log(B + 1))$ time and $O(B + K)$ working storage, rather than $\Theta(BK)$ direct score evaluations, for K options.

Weak duality makes the interval minimax value a valid upper bound on every feasible fixed-threshold LP in the interval; strong LP duality gives equality on feasible singleton intervals. A new epigraph-dual mapping proves that the exact minimax envelope bound is no larger than the bounded-threshold group-clique LP. Convex geometric bracketing and Lipschitz contraction return an evaluated valid bound together with an explicit minimization-error certificate. A best-first splitting algorithm then certifies the maximum LP value over the complete threshold family to a requested scaled tolerance.

The complete protocol and gates are serialized with the release. Across 60 primary instances, the compressed certificate reaches scaled tolerance 10^{-6} throughout, wins all 60 instance-level median comparisons, and is 2.33 times faster in geometric mean than a sparse bounded-threshold group-clique LP (95% design-stratified bootstrap interval 2.16–2.51). Speedup increases from 1.61 at 360 groups to 3.35 at 1,440 groups. All 36 robustness cases reach tolerance. A transparent two-option transformation of 9 published robust-knapsack coefficient sets wins 9 instance-level median comparisons with a 2.31-fold advantage; it is an out-of-generator stress test, not a replication of the source uncertainty model. Independent algebraic, epigraph-LP, and exhaustive-scan checks find maximum error 5.59×10^{-8} , with maximum scaled multiplier-certificate gap 1.00×10^{-8} .

The released implementation also embeds either interval bound in an exact integer threshold search. A separate audit confirms objective agreement and valid gap accounting, while showing that integer subproblem work can erase the oracle advantage and that compact SCIP remains preferable on many-breakpoints instances. The analysis therefore concerns simultaneous all-threshold Lagrangian evaluation for exactly-one groups, not the classical threshold reduction, a new robust formulation, or universal integer-solver superiority.